

Assignment 5.

1. Create a reducer function in a new file called `playlistReducer.js` that manages an array of songs, handling 'ADD_SONG' and 'REMOVE_SONG' actions using a switch case.
2. Build a simple Spotify-style playlist manager using `useReducer` in React — display a list of songs, and allow the user to add a new song by typing a name and clicking an 'Add' button.
3. Refactor your playlist manager so that each song in the list has a 'Delete' button next to it; dispatch a 'REMOVE_SONG' action from the Delete button in the child `SongItem` component.
4. Extend your reducer to handle an 'EDIT_SONG' action: allow users to click an 'Edit' button on a song, change its name in an input box, and save it by dispatching from the child component.

Hint: Pass the dispatch function as a prop to the child component that handles editing.

Answer:-

1 question :-

`PlaylistReducer.js`

```
const playlistReducer = (state, action) => {
  switch (action.type) {
    case "ADD_SONG":
      return [
        ...state,
        {
          id: Date.now(),
          name: action.payload,
        },
      ];

    case "REMOVE_SONG":
      return state.filter(
        (song) => song.id !== action.payload
      );

    case "EDIT_SONG":
      return state.map((song) =>
        song.id === action.payload.id
          ? {
              ...song,
              name: action.payload.name,
            }
          : song
      );
  }
};
```

```

        default:
            return state;
        }
    };

export default playlistReducer;

```

Question2:- PlatlistManager.js

```

import React, { useReducer, useState } from "react";

import playlistReducer from "../playlistReducer";
import SongItem from "../SongItem";

const PlaylistManager = () => {
    const [songName, setSongName] = useState("");

    const [songs, dispatch] = useReducer(playlistReducer, [
        {
            id: 1,
            name: "Shape of You",
        },
        {
            id: 2,
            name: "Perfect",
        },
    ]);

    const handleAddSong = () => {
        if (!songName.trim()) return;

        dispatch({
            type: "ADD_SONG",
            payload: songName,
        });

        setSongName("");
    };

    return (
        <div>
            <h1>🎵 Spotify Playlist Manager</h1>

            <input
                type="text"
                placeholder="Enter Song Name"
            />

```

```

        value={songName}
        onChange={(e) => setSongName(e.target.value)}
      />

      <button onClick={handleAddSong}>Add Song</button>

      <hr />

      {songs.map((song) => (
        <SongItem key={song.id} song={song} dispatch={dispatch} />
      ))}
    </div>
  );
};

export default PlaylistManager;

```

Question 3, 4:-

SongItem.jsx

```

import React, { useState } from "react";

const SongItem = ({ song, dispatch }) => {
  const [isEditing, setIsEditing] = useState(false);

  const [newName, setNewName] = useState(song.name);

  const handleSave = () => {
    dispatch({
      type: "EDIT_SONG",
      payload: {
        id: song.id,
        name: newName,
      },
    });

    setIsEditing(false);
  };

  return (
    <div
      style={{
        margin: "10px 0",
      }}
    >

```

```

    {isEditing ? (
      <>
        <input
          type="text"
          value={newName}
          onChange={(e) => setNewName(e.target.value)}
        />

        <button onClick={handleSave}>Save</button>
      </>
    ) : (
      <>
        <span>{song.name}</span>

        <button onClick={() => setIsEditing(true)}>Edit</button>

        <button
          onClick={() =>
            dispatch({
              type: "REMOVE_SONG",
              payload: song.id,
            })
          }
        >
          Delete
        </button>
      </>
    )}
  </div>
);
};

export default SongItem;

```